

🧠 System 2: Deliberation

Cadence: 0.5–2 Hz (500–2000 ms)

Substrate: Cloud / Host Linux MPU

Function: Semantic parsing, open-vocabulary 3D grounding, expiring intent leases (t_{expire}).

OpenVLA · LLaVA-OneVision · GPT-4o

🏗️ System 1.5: Planning

Cadence: 15–50 Hz (20–66 ms)

Substrate: Embedded SoC NPU / GPU

Function: Trajectory generation, multi-step action chunking ($H = 16$), temporal ensembling.

Diffusion Policy · ACT · Flow Matching

🛡️ System 1: Reflex

Cadence: 1000 Hz (1 ms)

Substrate: Bare-Metal / FreeRTOS MCU

Function: Real-time CBF invariants, spline evaluation, dynamic stopping veto, PWM timer registers.

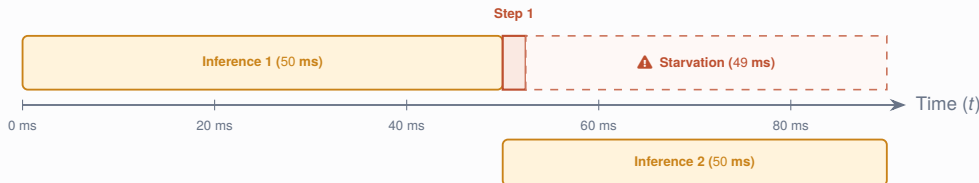
ARM Cortex-M4/M7 · STM32 · CBF-QP

3D Lease (t_{expire})

Chunk A_t ($H = 16$)

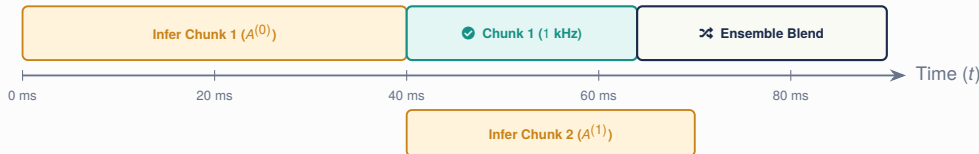
🕒 Latency Starvation vs. Action Chunk Delay Amortization

A. Single-Step Markovian Policy ($H = 1$) $\Rightarrow$ Discrete Latency Starvation & Mechanical Chatter



Gearbox Damage:
Motor jerks 1 ms, freezes 49 ms, exciting structural resonance.

B. Action Chunking ($H = 16, K = 8$) $\Rightarrow$ Delay Amortization & Continuous 1 kHz Trajectory



Zero Starvation:
Fluid C^2 motion via overlapping chunk horizons.